

I. Linear Algebra

Practice Problem 1. Involutions

Let V be a finite-dimensional real vector space and let $T : V \rightarrow V$ satisfy

$$T^2 = I.$$

For $v \in V$, define

$$v_+ = \frac{1}{2}(v + Tv), \quad v_- = \frac{1}{2}(v - Tv).$$

- Show that $v = v_+ + v_-$.
- Show that $Tv_+ = v_+$ and $Tv_- = -v_-$.
- Show that the eigenspaces corresponding to 1 and -1 intersect only at 0.
- Conclude that T is diagonalizable.

$$\begin{aligned} a) \quad v_+ + v_- &= \frac{1}{2}(v + Tv) + \frac{1}{2}(v - Tv) \\ &= \frac{1}{2}v + \frac{1}{2}Tv + \frac{1}{2}v - \frac{1}{2}Tv \\ &= v \end{aligned}$$

$$\begin{aligned} b) \quad Tv_+ &= T\left(\frac{1}{2}v + \frac{1}{2}Tv\right) = \frac{1}{2}Tv + \frac{1}{2}T^2v = \frac{1}{2}Tv + \frac{1}{2}v \\ &\quad \text{by linearity of } T \quad \text{by } T^2 = I \\ &= \frac{1}{2}(Tv + v) = \frac{1}{2}(v + Tv) \end{aligned}$$

$$Tv_- = \frac{1}{2}(v - Tv) \text{ by same steps}$$

(1) eigenspaces defined by basis of eigenvectors, all vectors x obey $Tx = \lambda x$ & $T^*x = \lambda^*x$

so if vector x corresponds to $\lambda = 1$:

$$Tx = x \Rightarrow T = I$$

to $\lambda = -1$:

so all vectors take

to $T = -I$:

$$T^n x = (-1)^n x$$

s.t. all vectors take form of V_+ by (b)

$$T x = x \text{ for } n \text{ even} \Rightarrow T = I$$

$$T x = -x \text{ for } n \text{ odd} \Rightarrow T = -I$$

s.t. eigenspaces take either V_+ or V_- by (b)

Since $\forall v \in V, v = v_+ + v_-$,

Thus the bases formed by eigenvectors

form basis for all of V ,

s.t. eigenspaces are linearly ind

to span entire space,

thus only intersect at 0

d) Some eigenvectors are linearly ind,
they form invertible X s.t.

$$T = X D X^{-1}, \text{ thus } T \text{ diagonalizable}$$

check here
e)

Practice Problem 2. A nilpotent operator

Let $V = P_4(\mathbb{R})$ and define $D: V \rightarrow V$ by $D(p) = p'$.

(a) Write the matrix of D in the basis $(1, x, x^2, x^3, x^4)$.

(b) Find $\ker D$ and $\text{range } D$.

(c) Explain why $D^5 = 0$.

(d) Show that $I + D$ is invertible and find $(I + D)^{-1}$ as a polynomial in D .

(e) Show similarly that $I - 2D$ is invertible and find its inverse.

4)

basis β

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \\ x^4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2x \\ 3x^2 \\ 4x^3 \end{bmatrix}$$

$$4) \quad D = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \text{basis } V \\ \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \\ x^4 \end{bmatrix} \end{matrix} = \begin{bmatrix} 0 \\ 1 \\ 2x \\ 3x^2 \\ 4x^3 \end{bmatrix}$$

$y \in V = P_4(\mathbb{R})$
 coefficients polynomial deg ≤ 4
 5×1

like $1 + x + x^2 + x^3 + x^4$

$$\rightarrow 0 + 1 + 2x + 3x^2 + 4x^3$$

$$b) \ker D = \{v \in V : D(v) = 0\}$$

thus all polynomials of deg ≤ 4

w/ derivative 0,

so all polynomials of deg 0,

constant terms

$$\{v \in V : (v, 0, 0, 0, 0), v \in \mathbb{R}\} \\ = \text{span}\{1\}$$

range:

all polynomials of deg ≤ 3

$$\{v \in V : (v_1, v_2, v_3, v_4, 0), v_i \in \mathbb{R}\}$$

c) D^5 would require taking derivative 5 times,
 since max degree polynomial in V is 4,
 5th derivative is 0
 range of D is always polynomials
 of one less degree

at one less degree
 since D strictly upper triangular w/
 0 on diagonal, $D^5 = \text{zero matrix}$

$$d) \quad I + D = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\det I + D = 1 - (1 \cdot 0) + (2 \cdot 0) - (3 \cdot 0) + (4 \cdot 0)$$

$$= 1 \neq 0$$

so $I + D$ invertible

ask at 06:1

polynomial in D ?

$$(I + D)^{-1} = I - D + D^2 - D^3 + D^4$$

e) $I - 2D$ invertible?
 find inverse

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 0 \\ 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 1 & -6 & 0 \\ 0 & 0 & 0 & 1 & -8 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$\det = 1$ by same args

$$(I - 2D)(I - 2D)^{-1} = I$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

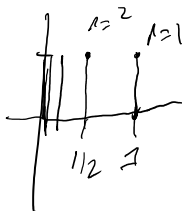
II. Topology and Continuity in $\mathbb{R}^n$

Practice Problem 3. A set with an accumulating edge

Let

$$S = \bigcup_{n=1}^{\infty} \left(\left\{ \frac{1}{n} \right\} \times [0, 1] \right) \subseteq \mathbb{R}^2.$$

- (a) Show that S is bounded but not closed.
- (b) Determine the closure $\overline{S}$.
- (c) Prove that your proposed closure is closed.
- (d) Is S compact? Is $\overline{S}$ compact?



$$a) \quad \forall s \in S : \left\{ \left(\frac{1}{n}, y \right) \in \mathbb{R}^2 \mid n \in \mathbb{Z}^+, 0 \leq y \leq 1 \right\}$$

bounded since by def.

$$0 \leq \frac{1}{n} \leq 1 \text{ and } 0 \leq y \leq 1$$

$$\text{so } \forall s \in S, s \subseteq [0, 1] \times [0, 1] \in \mathbb{R}^2$$

S contained in finite closed "ball"

not closed since the limit point $x_y^* = (0, y)$

is not contained in S for any

$y \in [0, 1]$, s.t. for all conv. seq.

$$x_{y_i} = \left\{ \frac{1}{n_i}, y \right\}_{i=1}^{\infty} \text{ as } n \rightarrow \infty, \lim_{n \rightarrow \infty} x_{y_i} = (0, y) = x_y^*$$

S fails to contain limit points so not closed

b) closure $\bar{S}$: let $C = \{(0, y) \mid \forall y \in [0, 1]\}$

$$\bar{S} = S \cup C = \left(\{0\} \cup \left\{ \frac{1}{n} \mid n \in \mathbb{Z}^+ \right\} \right) \times [0, 1]$$

c) $x_y^* \in \bar{S} \quad \forall y \in [0, 1]$ by construction of C ,

thus $\bar{S}$ contains all limit points

d) S not compact by Heine-Borel
since S not closed in $\mathbb{R}^2$,

$\Rightarrow$

since $\supset$ not closed in $\mathbb{R}$,
 $\bar{S}$ compact since closed &
 hence bounded for S

Practice Problem 4. Nested closed subsets of a compact set

Let $K \subseteq \mathbb{R}^n$ be compact and let

$$F_1 \supseteq F_2 \supseteq F_3 \supseteq \cdots$$

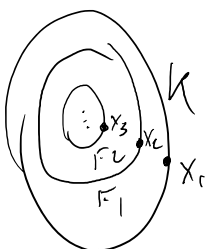
be nonempty closed subsets of K .

(a) Choose $x_n \in F_n$ and use compactness to obtain a convergent subsequence.

(b) Prove that

$$\bigcap_{n=1}^{\infty} F_n \neq \emptyset.$$

(c) Give a nested sequence of nonempty closed subsets of $\mathbb{R}$ with empty intersection, and identify the hypothesis that fails.



choosing
 $n) x_n \in F_n \quad \forall n \in \mathbb{Z}^+$

form a seq

$$\{x_1, x_2, x_3, \dots\} = \{x_n\}_{n=1}^{\infty}$$

Since $F_n \subseteq K$ for all F_n ,

all $x_n \in K$, this $\{x_n\}_{n=1}^{\infty}$
 is seq in compact K ,
 thus $\exists$ convergent subseq.
 $\{x_{n_i}\}_{i=1}^{\infty}$ by def compactness

b) since $\{x_{n_i}\}_{i=1}^{\infty}$ is convergent subseq,
 w/ $x_n \in F_n \forall n \in \mathbb{Z}^+$, $\lim_{i \rightarrow \infty} x_{n_i} = x_n^*$
 Since $F_1 \supseteq F_2 \supseteq F_3 \dots$
 $x_n^* \in F_n \forall F_n$
 Thus $x_n^* \in \bigcap_{n=1}^{\infty} F_n$
 $\neq \bigcap_{n=1}^{\infty} F_n = \emptyset$ since it must
 contain at least x_n^*

c) not compact? A2 is OK

Practice Problem 5. Positive separation

Let $K \subseteq \mathbb{R}^n$ be compact and $F \subseteq \mathbb{R}^n$ be closed, with $K \cap F = \emptyset$. Define

$$d(K, F) = \inf\{\|x - y\| : x \in K, y \in F\}.$$

- (a) Prove that $d(K, F) > 0$.
- (b) Give an example of two disjoint closed subsets of $\mathbb{R}^2$ whose distance is zero.

Hint for (a): If the infimum were zero, choose $x_n \in K$ and $y_n \in F$ with $\|x_n - y_n\| \rightarrow 0$.